

RS002 Week 4 — English Worksheet

Topic: Parallel Lists + Loops — A Full Pokédex · **Course:** Pokédex App · **Time:** about 45 minutes

This week your three starters grow into a **full Pokédex of 12+ Pokémon** across four parallel lists — name, type, HP, attack. Instead of writing a print line for each one, you use a **for loop** to walk through them all by index.

1 · Recall

From memory:

Write a list called `names` holding three Pokémon, then print the second one.

2 · Predict 🧠

Read each block. Before you run it, write what you think will happen.

```
names = ["피카츄", "이상해씨", "파이리"]
for i in range(len(names)):
    print(names[i])
```

`len(names)` is 3, so `range(len(names))` gives 0, 1, 2 . What three lines print?


```
names = ["피카츄", "이상해씨", "파이리"]
types = ["전기", "풀", "불"]

for i in range(len(names)):
    print(f"{names[i]} - {types[i]}")
```

The loop uses `i` in both lists. Write out every line it prints.


```
names = ["피카츄", "이상해씨", "파이리"]
hp = [35, 45]

for i in range(len(names)):
    print(f"{names[i]}: HP {hp[i]}")
```

`names` has 3 items but `hp` has only 2. What happens on the last loop? Why?

3 · Spot the Bug 🐛

Read what each block is **supposed** to do, fix it, then explain the fix.

Bug A — This should print **every** name in the list. Right now it only ever prints the first one.

```
names = ["피카츄", "이상해씨", "파이리"]
for i in range(len(names)):
    print(names[0])
```

Hint: the loop variable `i` changes each time — use it.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — A new Pokémon was added to `names` but not to the other lists. When the loop reaches the new one, the program crashes.

```
names = ["피카츄", "이상해씨", "파이리", "꼬부기"]
types = ["전기", "풀", "불"]

for i in range(len(names)):
    print(f"{names[i]} - {types[i]}")
```

Hint: parallel lists must all be the **same length**.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This should number the Pokédex starting at `#1`. Right now the first entry shows `#0`.

```
names = ["피카츄", "이상해씨", "파이리"]
for i in range(len(names)):
    print(f"#{i} {names[i]}")
```

Hint: the index starts at 0, but humans count from 1.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Modify It

Start from this working Pokédex:

```
names = ["피카츄", "이상해씨", "파이리", "꼬부기"]
types = ["전기", "풀", "불", "물"]
hp = [35, 45, 39, 44]
attack = [55, 49, 52, 48]

print(f"\n=== 도감 ===")
print(f"등록된 포켓몬: {len(names)}마리\n")

for i in range(len(names)):
    print(f"#{i+1} {names[i]} 타입:{types[i]} HP:{hp[i]} 공격:{attack[i]}")
```

Make these changes one at a time and run after each:

1. Add one Pokémon — remember to add to **all four lists**.
2. Add the word **마리** after the HP so it reads **HP:35** → **HP:35**. (Try it; see how f-strings let you wrap text around values.)
3. Print only the **name and type** inside the loop, dropping HP and attack.

Write your changed / added lines here:

5 · Make It 📷

Build your own **full Pokédex of at least 12 Pokémon** across four parallel lists (name, type, HP, attack). Use a **for** loop to print them all as a numbered list. Show your **trainer name from earlier weeks** at the top.

When it works, send a **photo or video** on KakaoTalk, then explain what you did. Use these starters — write 4 to 6 sentences.

First, I made four lists called ...

My **for** loop walked through them by ...

`len(names)` told me ... so the loop ran ... times.

To add a new Pokémon I had to ...

The trickiest part was ...

A loop is better than writing 12 print lines because ...

6 · Record Your Walkthrough 🎥

Film your Pokédex printing. Teach it like the viewer is new. Try to use: **parallel list**, **for loop**, **index**, **len**, **same length**.

1. Run your program and show the full numbered Pokédex.
2. Read your `for` loop out loud and say what `i` does each time.
3. Explain why all four lists must be the same length.
4. Add one Pokémon live and show it appear at the bottom.

Plan what you will say here first:

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.